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Hamburg, [PLATZHALTER — Datum der Absendung]

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Application as Postdoctoral Researcher — Probabilistic Deep Learning for Urban Air Quality (AEON-UP) — Reference 1056 – 2026/KU 2

Dear Dr. Ramacher and Dr. Karl,

I am writing to apply for the postdoctoral position on probabilistic deep learning for urban air quality. What draws me to AEON-UP is the clause most advertisements leave out: the predictions are to come *with uncertainty estimates*. That question is not an abstract interest for me.

Since finishing my post-doc I have built and debugged a deep learning pipeline in PyTorch: converting a 15-layer transformer from ONNX, extracting its internal representations, and running batched inference on GPU. In the course of it I found two systematic errors in my own pipeline. Both produced entirely plausible output — smooth, well-shaped results that raised no exception and failed no test. One had already reached a published write-up before I caught it, and I corrected it publicly:

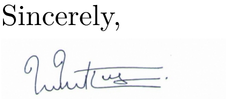
The lesson is the one AEON-UP is built around. A confidently wrong model is more dangerous than a visibly uncertain one, and where the output may inform decisions about what people breathe, a calibrated statement of what the model does *not* know is not a refinement — it is the product. Sparse, irregular monitoring networks are exactly the setting where a model must interpolate and declare how far to trust the interpolation.

During my post-doc at Universität Hamburg I developed a plankton life cycle model within the ERGOM framework and ran hindcast and projection simulations for warming scenarios, working daily with gridded NetCDF output on Linux HPC systems. I therefore know what a physics-based transport model is, what its output looks like at scale, and where its biases tend to live — which matters when the task is to couple such a model with a learned one rather than replace it. I followed the Urban Air Quality View work Dr. Ramacher mentored with ECMWF, bridging regional CAMS products to urban scale with a learned model rather than more expensive simulation, with particular interest. Contributing on the learned side of that coupling is exactly what I want to do.

I should be straightforward about where I stand. I come to the probabilistic side as a builder rather than as someone with a publication record in it: Bayesian methods and neural processes are current reading and implementation for me, not past papers. I am closing that deliberately — in September I take up a confirmed place on the certified course “Deployable Data Analysis & AI Pipelines with HPC” at the Supercomputing-Akademie of the HLRS, following earlier HPC training at the Jülich Supercomputing Centre. What I offer now is the combination the project needs: someone at home inside a physics-based modelling group who has spent the past year in the internals of a neural network, finding out where its output could not be trusted.

Hereon is not unfamiliar to me: I worked at the centre as a guest scientist in 2025 on ecosystem modelling, and would be glad to return. I hold work authorisation for Germany, live in Hamburg, and am available for the October start. I would welcome the chance to discuss the project.

Sincerely,



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